

Diagnostic Post-Processing Techniques for Bearing Ball Faults (CWRU Dataset)

Background and Challenges in Ball Fault Diagnosis

Diagnosing **ball faults** in rolling element bearings via vibration signals is challenging due to the weak, intermittent nature of ball fault impacts and interference from other machine vibrations. In the Case Western Reserve University (CWRU) benchmark dataset, ball faults proved “*by far the most difficult to diagnose correctly*” ¹. Small defect impacts are often masked by noise, non-stationary modulation, or even unintended raceway faults ² ³. These factors can smear or obscure the characteristic **ball-spin frequency (BSF)** harmonics in spectral analysis. Therefore, after initial signal segmentation and basic feature extraction, advanced **post-processing techniques** are crucial to reliably reveal ball fault signatures across various fault sizes.

Key challenges include:

- **Low Signal-to-Noise:** Ball fault impulses can be weak and buried in noise or structural resonances. Traditional spectrum or time-domain analysis may not clearly expose BSF-related peaks in such cases ².
- **Random Modulation Effects:** Ball faults may exhibit random amplitude modulations (e.g. due to slip or looseness) that smear fault frequency peaks ⁴ ⁵, making straightforward FFT peaks less obvious.
- **Coexisting Faults:** The CWRU data hints some bearings had simultaneous inner/outer race flaws (e.g. brinelling) alongside ball faults ⁶ ⁷. This can introduce spurious pulses at other fault frequencies (BPFI/BPFO) and mislead diagnosis if not properly filtered out.

Given these challenges, researchers have developed **post-processing methods** – applied after the initial segmentation and feature computation – to enhance true fault signatures and enable accurate classification. Below, we detail the most effective techniques (implementable with Python/SciPy) for ball fault diagnosis, and their performance on CWRU or similar benchmarks.

Envelope Analysis with Hilbert Demodulation

Envelope analysis is a foundational technique for bearing fault detection. It involves bandpass filtering the vibration signal around a high-frequency band where bearing impacts excite resonances, then using the **Hilbert transform** to extract the envelope (amplitude modulation) of that filtered signal. The envelope's spectrum is examined for periodic components at fault characteristic frequencies (BSF for ball faults, or its harmonics) ⁸ ⁹. This demodulation is effective because repetitive impact series (even if buried in noise) produce amplitude-modulated high-frequency resonance, which the envelope spectrum can reveal as distinct peaks.

In practice, one chooses a band containing the anticipated fault impact content (often determined by trial or via spectral analysis) and applies `scipy.signal.hilbert` to get the analytic signal and envelope. Envelope analysis was the first-line method (“Method 1”) in a comprehensive CWRU benchmark study ¹⁰, and it successfully highlighted most moderate-to-severe bearing faults via clear fault-frequency peaks ¹¹. For example, outer-race fault data showed **BPFO** harmonics with shaft-frequency sidebands in the envelope spectrum, enabling correct diagnosis ¹² ¹³.

However, **for small ball faults or noisy conditions**, a simple envelope of the raw signal may not suffice. The CWRU study noted many ball fault cases where “*the envelope spectrum is dominated by other effects... not a result of the ball fault*”, leading to missed diagnoses ³ ⁵. Thus, envelope analysis often needs enhancement via additional signal processing before it can reliably detect *BSF* for all fault sizes.

Cepstrum Pre-Whitening (Noise Floor Flattening)

Cepstrum pre-whitening is a post-processing approach that aims to flatten the spectral content of a signal, thereby amplifying weak impulsive components relative to deterministic tones or resonance peaks. In practice, one computes the spectrum, takes the logarithm (or cepstrum) to identify prominent frequency peaks, then filters or equalizes the spectrum to remove those peaks (effectively “whitening” the signal’s frequency response). Applying envelope analysis after such whitening can make repetitive impulses (like bearing fault impacts) much more pronounced in the envelope spectrum ¹⁴ ¹⁵.

Randall and Smith’s benchmark on the CWRU data implemented cepstrum pre-whitening as “Method 2” and found it **significantly improved diagnostic accuracy** for ball faults ¹⁵. By removing strong non-fault spectral components (e.g. gear mesh, $2\times$ line frequency, or resonance modes), pre-whitening allowed the true BSF harmonics to emerge in the envelope. In one example, a ball fault case where the raw-signal envelope gave no clear BSF (classified as *N1: no fault identified* ¹⁶) was successfully revealed after cepstrum pre-whitening – the envelope showed smeared peaks at $2\times$ BSF, $4\times$ BSF, $8\times$ BSF indicating a ball fault (*P1: partial fault diagnosis*) ¹⁶. Overall, the study concluded that cepstrum pre-whitening plus envelope analysis was the **most effective single method**, yielding the highest count of correct diagnoses (classified as Y1/Y2 outcomes) across difficult CWRU cases ¹⁵.

From an implementation standpoint, SciPy can perform this by combining FFT (`np.fft.rfft`), smoothing or gating the log-spectrum to remove prominent lines, then reconstructing a whitened signal via IFFT. The **design implication** for the user’s pipeline is that a pre-whitening filter stage can be inserted before the existing frequency-domain logic. This would complement time-domain features by ensuring that subsequent envelope or spectrum analysis sees a “flatter” noise floor where fault impulses stand out. Notably, using this technique, **all ball fault cases in CWRU became diagnosable** when combined with other enhancements ¹⁷, underscoring its real-world value.

Spectral Kurtosis and Optimal Band Selection (Kurtogram)

Another powerful post-processing approach is using **spectral kurtosis (SK)** to automatically find the frequency band that maximizes impulsiveness of the signal. The idea, introduced by Antoni’s *kurtogram* ¹⁸, is to scan different band-pass filter windows and compute the kurtosis of the filtered signal’s envelope; the band with highest kurtosis is likely to contain the repetitive transient (fault impacts) with minimal noise interference ¹⁹. This helps target the specific resonance band excited by a bearing fault, rather than

relying on a broad or fixed band. SciPy can facilitate this by designing band-pass filters (e.g. Butterworth via `scipy.signal.butter/filtfilt`) across a range of center frequencies and computing kurtosis (`scipy.stats.kurtosis`) to pick the best band.

Spectral kurtosis-based band selection (sometimes via its refined metrics like *autogram*, *infogram*, *sparsogram*, etc.) is widely regarded as a **state-of-the-art technique for bearing fault detection** ⁹. It was employed as “Method 3” (the “benchmark” method) in the CWRU study, which closely trailed pre-whitening in overall performance ¹⁵. SK effectively enhanced faint ball fault signatures: by focusing on the optimal frequency range, it produced envelope spectra with clear BSF harmonics in many cases that were otherwise unclear ²⁰ ¹⁶. An Applied Sciences 2021 review noted that “*spectral kurtosis... is still the reference algorithm for band selection*” in bearing diagnostics ²¹, as it tends to yield reliable fault frequency indications when a proper band is chosen.

One caveat reported is that kurtosis can sometimes be **over-sensitive to spurious impulsive noise**, highlighting random spikes that are not part of a repeating fault pattern ²². This can lead to false optimizations on noise bursts rather than true fault impacts. Modern variants (e.g. using Gini index or autocorrelation-based kurtosis) address some robustness issues ²³ ²⁴. In practice, combining SK with engineering judgment (verifying that the selected band’s envelope spectrum indeed shows fault-frequency peaks) is recommended.

Integration into the pipeline: The kurtogram approach can be added as a module that takes a segmented signal, performs band search, and outputs an optimal envelope spectrum or features (like peak amplitude at BSF, kurtosis value, etc.). This can directly complement existing frequency-domain logic by replacing a broad FFT with a targeted envelope analysis focused on the band of interest. The result is a more **accurate fault indicator** that works across different fault sizes: even small ball defects can be detected if their slight impacts are amplified by filtering in the right frequency band.

Wavelet Transform and Advanced Signal Deconvolution

Time–frequency techniques like the **wavelet transform** offer another route for post-processing, providing adaptive filtering of vibration signals. Wavelet analysis can isolate transient bursts at multiple scales, which is useful for bearing faults that produce brief, periodic pulses. By decomposing the signal (e.g. via `pywt` or `scipy.signal.cwt`) and either selecting the most informative detail coefficients or reconstructing a denoised signal, one can enhance fault-related transients. For example, a multi-scale principal component analysis with wavelets was shown to significantly increase the kurtosis of bearing fault signals, making faults more detectable ²⁵ ²⁶. In essence, wavelet-based filtering can act like an automated band-pass, often *improving SNR for incipient faults* better than fixed filters.

Beyond wavelets, a class of **blind deconvolution** methods has emerged for bearing diagnosis. Techniques such as **Minimum Entropy Deconvolution (MED)**, **Maximum Correlated Kurtosis Deconvolution (MCKD)**, and their variants aim to undo the smearing effect of the transmission path, shortening the impulse response so that fault impacts appear as more distinct spikes. These filters maximize kurtosis or minimize entropy of the signal, thereby accentuating repetitive impulses from faults. Recent research (Di et al., 2024) applied several of these on a CWRU ball fault signal – e.g. MED, particle-optimized MED, MCKD, and a fuzzy entropy deconvolution (FEAD) – all of which **sharpened the impulse train and clarified the envelope spectrum** compared to the raw signal ²⁷ ²⁸. Figure 1 below conceptually illustrates how such

filters can reveal the fault frequency content by producing cleaner periodic spikes in the filtered signal's envelope.

Fig. 1: Example of advanced filtering on a CWRU ball fault signal. Techniques like FEAD, MED, MCKD, etc., produce filtered signals whose envelope spectra show clearer BSF harmonics, improving fault diagnosability (adapted from Di et al., 2024).

In practical terms, implementing MED/MCKD from scratch is non-trivial, but SciPy can approximate some effects via iterative filters or by using open-source toolboxes. The **design implication** is that incorporating a deconvolution stage (where the signal is processed to maximize some impulse indicator) can greatly aid in post-processing. It would slot into the pipeline before feature extraction – essentially as an *advanced preprocessing*, yielding a cleaner signal for the existing feature logic to work on. The payoff can be substantial: one study combining wavelet denoising with envelope entropy features achieved *over 99% classification accuracy* on CWRU and similar datasets ²⁹. These methods are thus promising for ensuring that even subtle ball faults (e.g. 0.007" diameter) are not overlooked.

Feature Fusion and Machine Learning Classification

In addition to signal-processing techniques, **data-driven post-processing** can dramatically improve fault diagnosis accuracy. After extracting time and frequency features from each signal segment (as the user's pipeline already does), an effective strategy is to feed these features into a **machine learning classifier** that learns to distinguish fault types (ball vs inner race vs outer race vs normal). Traditional time-domain features (RMS, kurtosis, crest factor, etc.) combined with frequency-domain indicators (e.g. spectral peak magnitudes near BSF, BPFI, BPFO) provide a descriptive feature vector. A supervised classifier like a Support Vector Machine (SVM), Random Forest, or a shallow neural network can be trained (using labeled examples from the CWRU dataset) to recognize the pattern of features corresponding to a ball fault. This kind of post-processing moves from hard-coded thresholds to a learned decision function, often boosting accuracy by capturing subtle combinations of features that indicate a fault.

For instance, a recent study extracted 24 features (15 time-domain, 5 frequency-domain, 4 nonlinear chaotic features) and then used a feature selection + LSTM-attention classifier. It achieved **98.74% F1-score on the CWRU dataset** (10-class fault identification) with just 8 optimized features ³⁰. This demonstrates that even a relatively compact feature-based model can nearly match deep learning performance on this benchmark. In a simpler approach, one could train an SVM on the same features – prior works have reported high accuracy (often >95–99%) in classifying CWRU faults using standard ML algorithms as well ³¹ ³². The benefit of such classifiers is that they naturally integrate with the existing pipeline: the features computed in the time/frequency domain are used as inputs, so the current logic (feature extraction) remains useful. The **extension** is to include a classification stage that outputs a fault diagnosis (with confidence) instead of or in addition to rule-based alarms.

Modern deep learning models go even further, by performing end-to-end feature learning from raw signals or spectrograms. Convolutional neural networks (CNNs) and others have been extensively tested on CWRU – often achieving **99–100% accuracy** in distinguishing fault categories under laboratory conditions ³² ³³. For example, 1-D CNNs on raw vibration segments have reached ~99.9% accuracy on 10-class CWRU tasks ³², and 2-D CNN approaches (treating the vibration signal as an image or spectrogram) have even achieved 100% in some cases ³⁴ ³⁵. These results illustrate the *ceiling* of performance on this dataset when abundant training data and deep models are used. However, a deep learning solution would

represent a more significant change to the pipeline (requiring network training and potentially more computational resources). If the goal is to **complement the existing time/frequency logic**, a pragmatic choice is a lightweight ML model using the features already available, which can still deliver near-optimal accuracy.

Comparison of Techniques and Recommendations

The table below summarizes the discussed post-processing techniques, their approach, and reported effectiveness on ball fault diagnosis (especially referencing CWRU outcomes):

Technique	Approach & Integration	Performance on CWRU (benchmark studies)
Hilbert Envelope Analysis	Demodulate a filtered high-frequency band to expose periodic impacts (uses Hilbert transform for envelope). Can be added after initial FFT to highlight BSF harmonics.	Baseline method – identified most clear faults but struggled on small faults or noisy cases ¹ (envelope peaks sometimes masked by other modulation ³). Useful as a first step, but not fully reliable alone for all ball fault sizes.
Cepstrum Pre-Whitening + Envelope	Flatten the spectrum (remove strong deterministic components) then apply envelope analysis. Easily done with FFT->log->IFFT in SciPy. Integrates as a pre-processing filter before envelope/feature extraction.	Top performer in benchmarks: Delivered the most correct diagnoses for difficult cases ¹⁵ . For example, enabled detection of ball fault harmonics where raw signal analysis failed ¹⁶ . In combination with other methods, achieved 100% detection of all fault cases in CWRU ¹⁷ .
Spectral Kurtosis (Kurtogram)	Compute kurtosis of filtered sub-bands to automatically select an optimal band for envelope analysis. Can be implemented as a scanning algorithm in Python. Integrates as an intelligent filter stage.	Highly effective: Considered a gold standard for bearing diagnostics ²¹ . Reveals fault frequency lines even under heavy noise if the right band is chosen. In CWRU tests, SK-based filtering correctly identified many faults missed by broad-band analysis. Slight sensitivity to random impulsive noise noted ²² , but overall a robust method for ball fault feature enhancement.
Wavelet & Deconvolution	Use wavelet transform for multi-scale denoising or apply blind deconvolution (MED, MCKD, etc.) to compress impulses. This stage would precede feature extraction, outputting a “cleaned” signal emphasizing transients.	Noise-robust, improves small fault detection: Research combining wavelet shrinkage with envelope entropy achieved >99% classification accuracy on CWRU ²⁹ . Deconvolution methods (MOMEDA, MCKD) have shown clear improvements in extracting BSF signatures from noisy signals ²⁷ ²⁸ . These approaches ensure even subtle faults become pronounced in the processed signal.

Technique	Approach & Integration	Performance on CWRU (benchmark studies)
Machine Learning Classifiers	Fuse time-domain, frequency-domain, and possibly envelope features into a classifier that outputs fault type. Can be a simple ML model (SVM/RF) or a neural network, added at the end of the pipeline.	State-of-the-art accuracy: With sufficient training data (like CWRU), feature-based ML can reach ~98–99% accuracy in fault identification ³⁰ . Deep learning models (if integrated) can hit ~99.9% ³² or even 100% on this dataset ³⁴ . This data-driven stage greatly enhances reliability and consistency of diagnosis across fault sizes and conditions.

Recommendations: For the user’s diagnostic module (`dx1.py`), which already computes time and frequency features, the most practical high-impact improvements would be:

- **Incorporate Envelope Spectrum Analysis** (Hilbert demodulation) focused on the expected ball fault band. This will complement the existing time-domain indicators by directly targeting the fault frequency evidence. It can be triggered after segmentation on each signal frame and provide features like envelope peak at BSF or overall envelope kurtosis.
- **Add a Pre-Processing Filter:** Implement cepstral pre-whitening or an SK-based band-pass filter prior to the envelope extraction. This step will enhance weak fault signals. For example, one could use SciPy to perform a high-pass or band-pass around the bearing’s natural frequency (where ball fault impacts typically manifest) and/or apply a whitening filter using the signal’s cepstrum. This is relatively straightforward with Python and can dramatically improve the clarity of features used in the next steps.
- **Employ a Learning Algorithm for Decision:** Rather than relying on fixed thresholds from time/frequency logic, train a lightweight classifier using historical labeled data (which CWRU provides amply). This classifier can ingest the features (RMS, kurtosis, peak spectrum amplitudes, etc., possibly augmented by envelope features from the above steps) and output a probabilistic diagnosis (e.g. “ball fault” vs “normal” vs other fault). This makes the pipeline adaptive – the system learns the complex feature patterns of ball faults, including how they differ for small vs large fault sizes, instead of manual tuning for each case. Given the high separability of CWRU fault data, even a modest ML model should considerably boost accuracy and consistency.
- **Design for Modularity and Fusion:** The pipeline extension should be modular – e.g. a function for “enhanced envelope analysis” and a module for “classification decision”. This design allows testing the new components independently and also enables **fusion** with existing logic. For instance, the system could cross-verify: if the rule-based logic flags a possible ball fault and the ML classifier agrees with high confidence, then the diagnosis is very robust. If they disagree, the new envelope feature results might be used to adjudicate. Over time, reliance can shift more to the data-driven outputs as they are validated.

In summary, the **best-performing approach** for ball fault diagnosis is a *hybrid strategy*: advanced signal processing (pre-whitening, optimal filtering, or wavelet de-noising) to expose the fault’s spectral signatures, followed by a learned classification or robust decision rule that leverages both time-domain stats and frequency-domain/envelope features. This approach has been shown to achieve nearly perfect accuracy on

benchmark data like CWRU ²⁹ ³², and it addresses the weaknesses of any single method alone. By integrating these post-processing techniques, the diagnostic pipeline will be more sensitive to small faults, more immune to noise or confounding vibrations, and ultimately more **accurate** and **reliable** in real-world bearing fault detection.

References: The recommendations above are supported by extensive research and benchmark evaluations. For example, Smith & Randall's study of CWRU data demonstrated the effectiveness of cepstrum pre-whitening and spectral kurtosis methods ¹⁵ ²¹, while numerous machine learning studies have reported >98–99% accuracy on this dataset using advanced feature processing ³⁰ ³². These methods collectively ensure that ball fault diagnosis can be performed with a high degree of confidence across varying fault sizes and operating conditions.

¹ ² ³ ⁴ ⁵ ⁶ ⁷ ⁸ ¹⁰ ¹¹ ¹² ¹³ ¹⁴ ¹⁵ ¹⁶ ¹⁷ ²⁰ ²² (PDF) Rolling Element Bearing Diagnostics Using the Case Western Reserve University Data: A Benchmark Study

https://www.researchgate.net/publication/276248554_Rolling_Element_Bearing_Diagnostics_Using_the_Case_Western_Reserve_University_Data_A_Benchmark_Study

⁹ ¹⁸ ¹⁹ ²¹ ²³ ²⁴ Performance of Envelope Demodulation for Bearing Damage Detection on CWRU Accelerometric Data: Kurtogram and Traditional Indicators vs. Targeted a Posteriori Band Indicators

<https://www.mdpi.com/2076-3417/11/14/6262>

²⁵ ²⁶ Bearing fault analysis using kurtosis and wavelet based multi-scale PCA

<https://www.extrica.com/article/20560/pdf>

²⁷ ²⁸ The filtered signals and their envelope spectra from the CWRU bearing... | Download Scientific Diagram

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²⁹ The filtered signals and their envelope spectra from the CWRU bearing... | Download Scientific Diagram

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³⁰ Accuracy of the proposed method under different learning rates on the... | Download Scientific Diagram

https://www.researchgate.net/figure/Accuracy-of-the-proposed-method-under-different-learning-rates-on-the-CWRU-and-XJTU_fig5_369786439

³¹ ³² ³³ ³⁴ ³⁵ (PDF) Bearing Fault Detection and Diagnosis Using Case Western ...

https://www.researchgate.net/publication/340976229_Bearing_Fault_Detection_and_Diagnosis_Using_Case_Western_Reserve_University_Dataset_With_Deep_Learning_Approaches_A_Review